# Introduction

Research suggests that there is a relationship between hormonal patterns across the span of a day and conditions including sleep disorders and depression. In this study, the focus is on the hormones, DHEA and Cortisol, which play a role in regulating stress. Cortisol is an important steroid hormone that significantly affects metabolism which in turn affects the central nervous system. DHEA is

In this study, researchers followed the cortisol and DHEA patterns over the course of a day for a totstudal y period of three days to measure the the patterns of the two hormones in a natural environment.
Researchers utilized two methods to collect saliva samples; a Saliva Procurement and Integrated Testing (SPIT) booklet and an automatic Electronic Monitoring Cap. Saliva samples were collected at waking, 30 minutes after waking, before lunch, and 600 minutes after waking. Subjects did not have to adhere to a specific wake up time for 31 healthy control subjects.

This report addresses the following hypotheses of interest:

**Hypothesis 1:** There was satisfactory agreement between the subject's recording of sampling times compared to the recording times by the electronic monitoring cap. More specifically, there is a one-to-one relationship between the subject's recorded times and the booklet times.

**Hypothesis 2:** Subjects adhered appropriately to the study protocol of the +30 minute and +600 minute sampling times.

**Hypothesis 3:** DHEA and Cortisol levels show an initial increase from waking to 30 minutes followed by a gradual decline from 30 minutes and on.

*We will further refer to the electronic monitoring cap as MEM time and the the SPIT Booklet as Booklet time for the remainder of the report.*

# Methods

## Preliminary Data Analysis

Below is a Table1 of the primary variables of interest for the analysis:

Table 1: Characteristics of Study Cohort

| | Overall | Wake Time | +30 Min | Pre Lunch | +600 Mi |
|---|---|---|---|---|---|
| n | 372 | 93 | 93 | 93 | 93 |
| Minutes Since Waking (Midnight) (mean (SD)) | 235.78 (262.28) | 0.00 (0.00) | 33.49 (8.22) | 321.89 (84.42) | 642.09 (85 |
| Booklet Time (Min Since Midnight) (mean (SD)) | 654.35 (272.68) | 415.91 (86.85) | 448.09 (85.46) | 736.34 (71.54) | 1056.89 (11 |
| MEM Time (Min Since Midnight) (mean (SD)) | 684.24 (279.26) | 417.58 (88.41) | 463.51 (89.78) | 734.99 (68.44) | 1068.24 (12 |
| DHEA (nmol/L) (mean (SD)) | 0.98 (1.03) | 1.78 (1.27) | 1.11 (0.92) | 0.51 (0.46) | 0.50 (0.6 |
| Cortisol (nmol/L) (mean (SD)) | 5.95 (6.95) | 7.38 (6.05) | 9.00 (4.94) | 3.21 (2.29) | 4.15 (10.3 |

For analysis, we exclusively worked with the time variables in minutes. Looking at Minutes Since Waking, we can see that the mean minutes closely follow the study protocol times. The average DHEA and Cortisol levels appear to be similar across the 4 time points throughout the day.

As missingness can present serious issues for the data analysis if not addressed, we investigated missingness through calculating the number and percentage of missing values for each variable in the dataset. We further visualized missingness for each variable.

For the variables we employed in our analyses, we assessed and visualized distributions to determine if there were any outliers or considerations that needed to be taken for data analysis.

## Data Cleaning

For DHEA, we removed observations that were $\geq 5.205$ (nmol/L) as these are considered extremely high DHEA levels that could interfere with analysis. There were 6 observations (4 of which corresponded to Subject 3037) that had DHEA levels of 5.205 (nmol/L). We removed all observations for Subject 3037 due to multiple high DHEA values that suggest underlying health problems as well as the 2 other observations with DHEA of 5.205 (nmol/L). For cortisol, we observed 5 observations from 4 patients that had Cortisol levels $\geq 26$. Although these values are considered high, they are biologically plausible and they were left in the analysis. One observation had a cortisol level of ~89 which was removed for the analysis.

## Data Analysis

To address the first hypothesis, we employed a longitudinal linear mixed effects model with Booklet Time in Minutes as the outcome with covariates - MEM in Minutes and the Day Number (1-3). Adding Day as a covariate noticeably improved diagnostic plots for the assumptions of heteroscedasticity and normality. We included a random intercept for subject that allowed us to take variability between subjects into account in our model. To assess whether booklet times tended to be recorded earlier or later than MEM time, we used the difference between MEM and Booklet times as the outcome in a longitudinal linear mixed effects model. This model had Day Number as a covariate and a random intercept for subject.

To determine whether subjects were adhering accurately to the +30 minute and +600 minute sampling times required by study protocol, intervals of time were defined to calculate percentages for which there was good or adequate adherence. Good adherence was defined by a collection of the saliva sample within a 7.5 minute window interval. Patients had adequate adherence if samples were within a 15 minute window. To calculate the percentage of individuals that had good adherence and adequate adherence, the difference between the "correct" sampling time and the observed time was calculated and this difference was used to assess whether observed times were within the good or adequate adherence windows. We visualized and summarized the distribution of the MEM and Booklet times to guide us in our decision to proceed with either MEM or Booklet times for our calculations. We observed that MEM times had a considerable amount of missingness and determined that reporting both would give a better overall sense of adherence. We summarized adherence for the study population with tables.

For the third hypothesis, we used a piecewise longitudinal linear mixed effects model to assess whether DHEA and Cortisol levels followed the expected diurnal pattern. Due to a right-skew in the distribution of DHEA, a log transformed DHEA variable was used as the outcome to assess DHEA changes over the day. No transformations was made to Cortisol. Covariates for both models included manually calculated pre and post 30 minute variables (variables for piecewise slopes) which would give us the slopes for the two time intervals of interest.

All analyses were conducted in R version 4.4.1 (2024-06-14), using the nlme package for modeling.

# Results

From the missingness investigation, we determined the MEMs had the highest amount of missing: 177 missing values which is equivalent to 47.6%. Booklet had the second highest amount of missingness with 55 missing values or 14.8%. Cortisol and DHEA measurements had 5 missing values each or 1.34% missingness.

## Hypothesis 1

Below is a table of the estimates, 95% Confidence Intervals and P-values:

Table 2: Model Summary

| | Value | Std.Error | DF | t-value | p-value |
|---|---|---|---|---|---|
| (Intercept) | 3.436 | 4.355 | 41 | 0.789 | 0.435 |
| MEM_C | 0.986 | 0.010 | 41 | 102.912 | 0.000 |
| factor(DAYNUMB)2 | -3.148 | 1.550 | 41 | -2.030 | 0.049 |
| factor(DAYNUMB)3 | -1.642 | 1.558 | 41 | -1.053 | 0.298 |

Holding day constant, a one-minute increase in MEM-recorded wake time is associated with an approximately one-minute increase in Booklet-recorded wake time ($\beta \approx 0.99$; 95% CI: (0.97–1.01); $p < 0.0001$). This near one-to-one association indicates strong concordance between the two wake-time measures within a given day. In other words, the subject's recorded wake up time and the MEM's recorded wake up time track each other quite closely. This is visualized in Figure 1 below. Changes in MEM time are mirrored almost exactly by changes in Booklet time and our initial hypothesis is correct. In regard to bias, booklet times tended to be recorded approximately 7-8 minutes earlier (95% CI: (-14.28, -0.62)) than MEM times on average. This difference is statistically significant with p-val $\approx$ 3.

## Hypothesis 2

Below we summarize the percentage of subjects that adhered to the study sampling times, for both MEM and Booklet times.

Table 3: Percentages for Good and Adequate Adherence by Collection Window

| Window | Method | Good | Adequate |
|---|---|---|---|
| +30 minutes | Booklet | 78.16 | 89.66 |
| +30 minutes | MEM | 52.86 | 71.43 |
| +60 minutes | Booklet | 46.25 | 56.25 |
| +60 minutes | MEM | 32.53 | 39.76 |

Overall, more subjects are meeting the adequate adherence thresholds compared to the good adherence thresholds for both the +30 minute and +600 minute sampling times. In

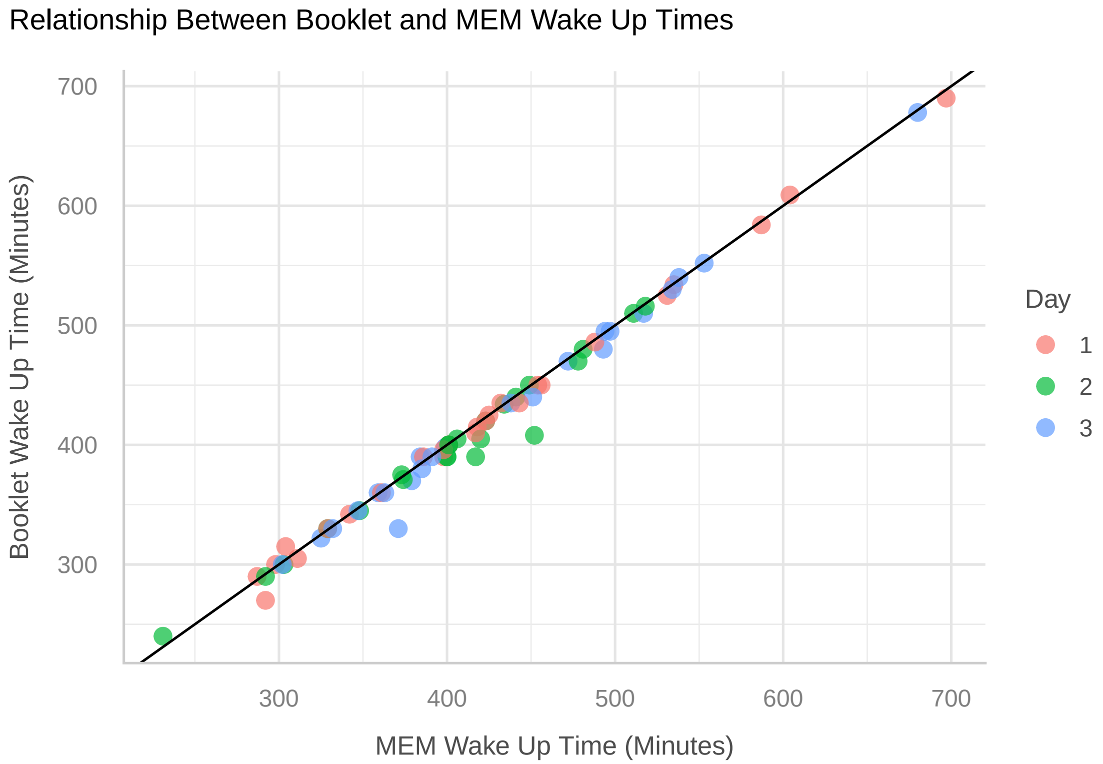


Figure 1: Booklet Vs. MEM Wake Up Times

addition, the MEM percentages are lower for both sampling times and for both good and adequaate thresholds.

In Figure 2 we visualize the minutes from the +30 minute sampling time (left column) and the +600 minute sampling time (right column). Times for Booklet are in the top row and Times for MEM are in the bottom row. The red dashed lines indicate the $\pm$ 7.5 window for good adherence while the pink dashed lines are the $\pm$ 15 minute window for adequate adherence.

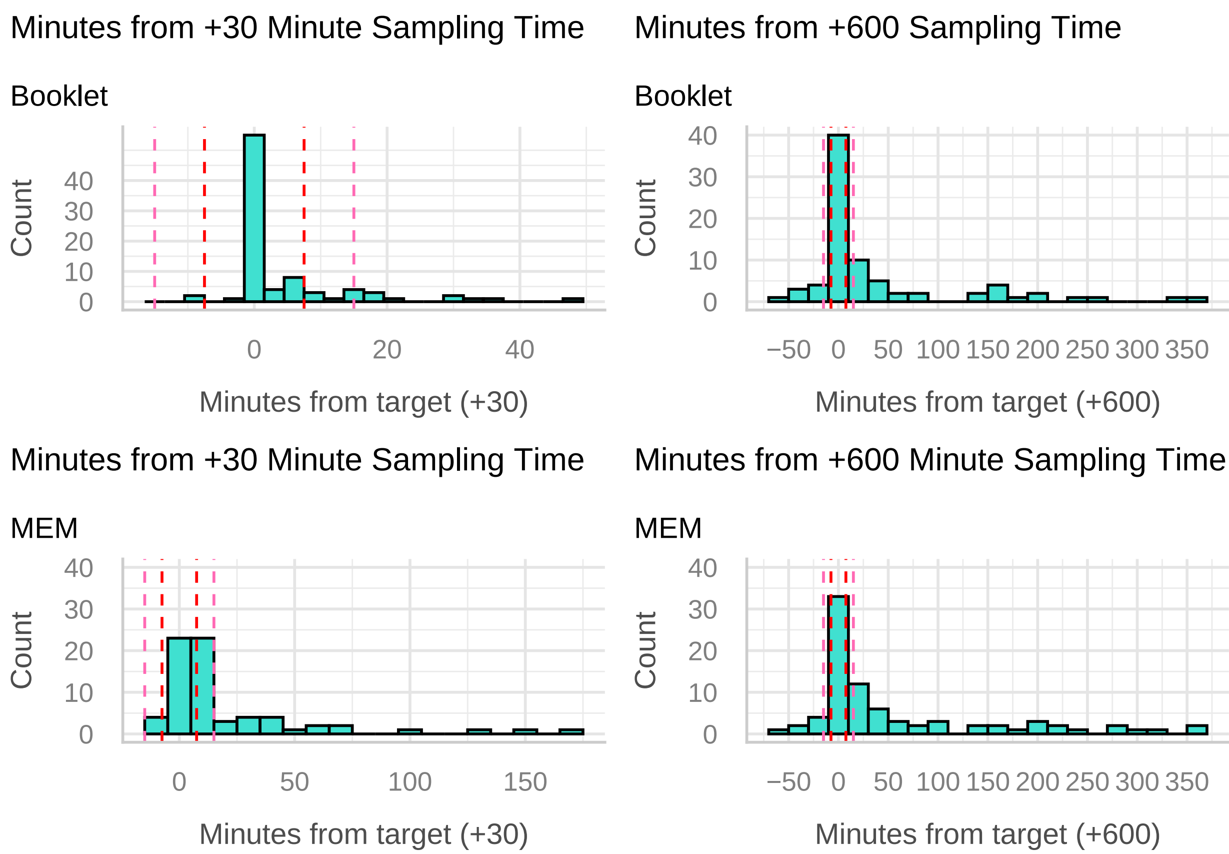


Figure 2: Adherence for +30 and +600 Sampling Times

## Hypothesis 3

### DHEA

In Table 3 we display the estimates, 95% Confidence Intervals and P-values for the DHEA model:

Table 4: Model Summary

| | Value | Std.Error | DF | t-value | p-value |
|---|---|---|---|---|---|
| (Intercept) | 0.21392 | 0.11741 | 288 | 1.82195 | 0.0695 |
| time_pre30 | -0.01724 | 0.00301 | 288 | -5.73083 | 0.0000 |
| time_post30 | -0.00153 | 0.00016 | 288 | -9.87455 | 0.0000 |

For the time period of wake time to 30 minutes, there is approximately a 40% decrease in

DHEA (95% CI: (-50.04%, -28.75%)). This is statistically significant with $p < 0.001$. DHEA shows a statistically significant decrease, not the expected increase, from waking to 30 minutes after waking.

As hypothesized, from 30 minutes to 600 minutes, there is approimately a 60% decrease in DHEA levels (95% CI: (-66.65%, -51.91%)). This is statistically significant with p-val < 0.001. We visualize these slopes in Figure 3 below.

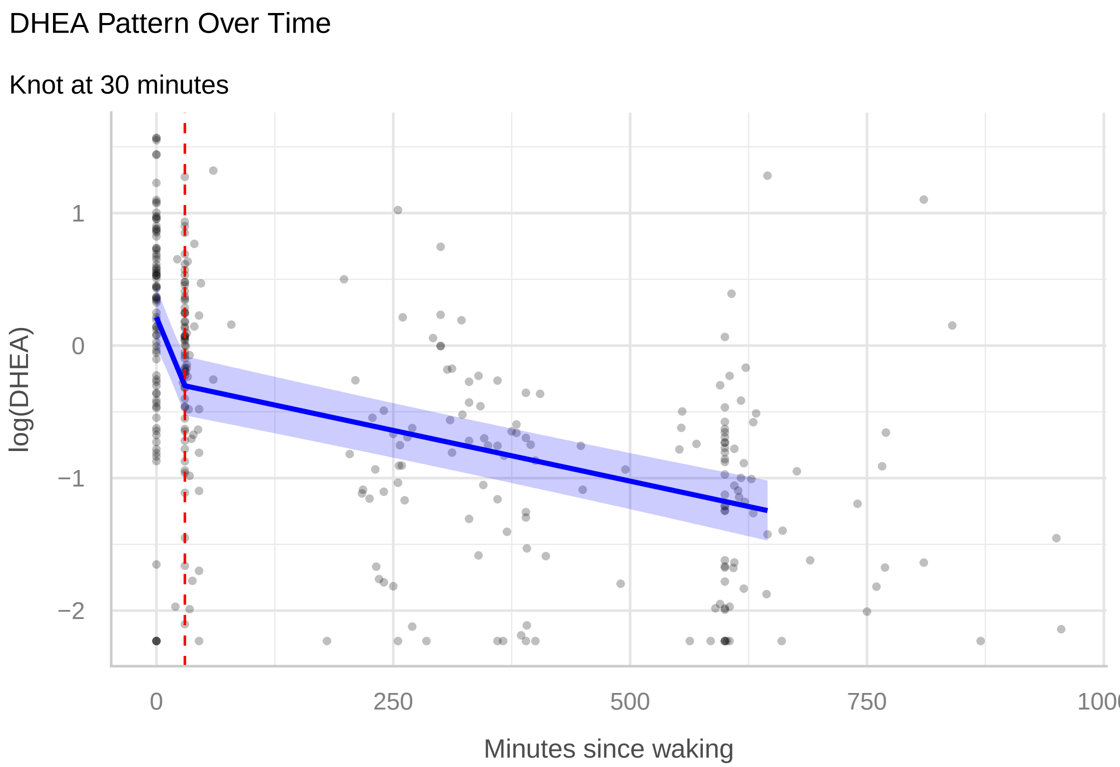


Figure 3: Diurnal DHEA Pattern

**Cortisol**

For the time period of wake time to 30 minutes, there is approximately a 54% increase in Cortisol (95% CI: (-60.19%, 503.15%)). This is not statistically significant with $p = 0.53$. Although there is an increase from waking to 30 minutes after waking, it is not a statistically significant increase.

Table 5: Model Summary

| | Value | Std.Error | DF | t-value | p-value |
|---|---|---|---|---|---|
| (Intercept) | 7.22161 | 0.56345 | 287 | 12.81668 | 0.00000 |
| time_pre30 | 0.02522 | 0.02295 | 287 | 1.09886 | 0.27275 |
| time_post30 | -0.00938 | 0.00119 | 287 | -7.91625 | 0.00000 |

From 30 minutes to 600 minutes, there is approimately a 99% decrease in DHEA levels (95% CI: (-99.86%, -97.27%)). This is statistically significant with p-val < 0.001. As expected, Cortisol showed a statistically significant decline over 10 hours. Cortisol follows the expected pattern of an increase (although not significant) from waking to 30 minutes followed by a decline from 30 minutes and on.

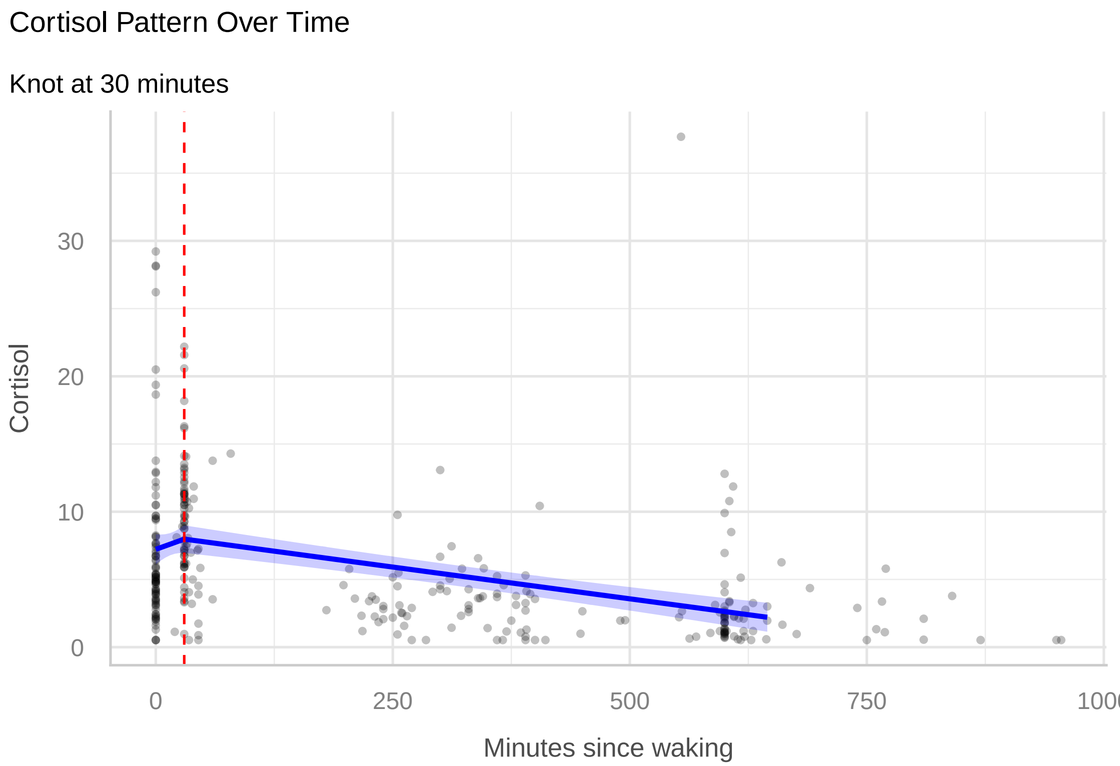


Figure 4: Diurnal Cortisol Pattern

## Discussion and Summary

In our missingness investigation, we determined that the variable with the highest missingness was MEM time. This was expected due to potential cap malfunction, such as low battery, or user error. In contrast, booklet-recorded time had much lower missingness. Measurements of DHEA and cortisol did not have substantial missingness, indicating that missingness was largely limited to the timing variables instead of the biological outcomes.

In regard to the comparison of MEM and booklet times, we found a statistically significant and strong linear relationship between the two measures, indicating that booklet-recorded times tracked closely with MEM times. However, we also observed that booklet times tended to be recorded slightly earlier than MEM times, on average. This suggests the presence of a small systematic bias. Possible explanations include delayed electronic cap recordings due to technical issues or perhaps inaccuracies in self-reported booklet times. While we are unable to directly determine the cause of this discrepancy, it highlights a known limitation of studies relying on self-reported sampling times.

For the +30 minute sampling window, the percentages for both good and adequate adherence were reasonably high, at approximately 80%. This indicates that participants were able to comply with the short-term post-waking sampling requirement for the most part. On the other hand for the +600 minute sampling window, we saw a notable decrease in adherence. Both good and adequate adherence percentages for booklet and MEM times fell below 50%. This decrease is expected, since longer intervals from waking are more susceptible to schedule variability, forgetfulness, and other daily activities, which undoubtedly can affect accurate timing. Across both sampling windows and adherence thresholds, the impact of missing MEM time was apparent, with lower adherence percentages compared to booklet time. This emphasizes the importance of appropriately addressing missingness in electronically recorded timing data, as differences between MEM and booklet times would likely affect downstream analyses and therefore potentially affect conclusions regarding adherence and hormone trajectories. In Figure 2, we observed clear differences between the distributions of minutes from the target sampling times for the +30 minute and +600 minute windows, as well as between MEM and booklet times. All distributions showed long right tails, indicating larger deviations from the target sampling times for some observations. However, the distributions were centered near zero, suggesting that the majority of subjects collected samples within the good or adequate adherence windows.

In the assessment of DHEA and cortisol patterns over time, we found that cortisol followed the expected diurnal pattern, with higher levels earlier in the day and a decline over time. In contrast, DHEA did not exhibit a clear or consistent pattern over the sampling period. These differences may be due to several factors, including biological differences in hormone regulation, variability within this particular cohort, or limitations in sampling frequency and timing.

In summary, our findings suggest that booklet-based saliva collection using the SPIT device demonstrates strong agreement with electronic monitoring, though a small systematic bias toward earlier recorded booklet times was observed. Adherence to short-term sampling requirements was generally high, while adherence to longer-term sampling windows was more variable. Cortisol exhibited the expected diurnal decline, whereas DHEA did not show a clear temporal pattern in this sample. These results highlight the feasibility and limitations of at-home saliva collection protocols and emphasize the importance of accurate timing and careful handling of missing data in studies of diurnal hormone patterns.